VERIFICATION REPORT
48 MW Houqiao Hydropower Project, Yunnan, China

Document Prepared By
Japan Consulting Institute

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Summary:

Japan Consulting Institute (JCI) has performed the verification of the project "48 MW Houqiao Hydropower Project, Yunnan, China", with regard to the relevant requirements of VCS Program. This verification covers the period from 1st April, 2006 to 30th November, 2011.

It is JCI's responsibility to express an independent verification statement on both the reported GHG emissions and the GHG emission reductions from the project activity for the period.

JCI's verification is based on the requirements as defined under the VCS Program Guide and VCS Standard, as well as the related VCS Guidance including the Validation and Verification Manual.

The verification was performed based on the following documentation:

1) Registered PD (ver.1.3) dated 14/10/2009 /1/.
2) Approved methodology ACM0002 (ver.10), "*Consolidated methodology for grid-connected electricity generation from renewable sources*" /11/.
3) Tool to calculate the emission factor for an electricity system, version 01.1 /14/
4) Monitoring Report version 01.1 dated 30/07/2012 /3/ and version 03.0 dated 15/03/2013 /4/

The verification process covered the auditing techniques to assess the quality of the information including desk review and on-site assessment, and also the quality confirmation of evidence in accordance with the VCS Validation and Verification Manual version 3.0 /18/.

JCI has verified that the information provided by the Project Proponent was actual and representative of current operation of the project activity, that it has been in accordance with the Monitoring Plan in the registered PD /1/ and the applied Methodology /11/, and that the emissions reduction achieved have been correctly determined.

In summary, it is JCI's opinion as the Validation/Verification Body that the information, described in the Monitoring Report version 03.0 /4/, meets all relevant VCS rules and requirements of the VCS Program and all relevant host country criteria. And also, based on the information confirmed and verified by JCI, it is JCI's opinion that 943,454 tonnes of CO_2 equivalents, indicated in the Monitoring Report version 03.0 /4/, is correctly determined as the Emission Reductions in accordance with the VCS Program during the period from 01st April, 2006 to 30th November, 2011. JCI thus requests the issuance of the Verified Carbon Units (VCUs).

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1. INTRODUCTION

The Project Proponent, Yunnan Baoshan Binglang River Hydropower Development Co., Ltd. has commissioned JCI to perform the verification of the “48 MW Houqiao Hydropower Project, Yunnan, China” (hereafter called “the project”) for the monitoring period from 1st April, 2006 to 30th November, 2011. This report summarizes the findings of the verification of the project, performed on the basis of all rules and requirements set out under VCS Program, Methodology and the related rules of UNFCCC.

1.1 Objective

Verification was conducted once the project implementation has commenced. It was the ex-post assessment of the monitored GHG data and information. During verification, JCI as Validation and Verification Body (VVB) evaluated the Monitoring Report and assess the following in accordance with Validation and Verification Manual of VCS Program /18/:

- The extent to which methods and procedures, including monitoring procedures, have been implemented in accordance with the validated Project Description (PD). This includes ensuring conformance with the monitoring plan.
- The extent to which GHG emission reductions and removals reported in the monitoring report are materially accurate.

1.2 Scope and Criteria

The scope of a verification helps place physical and temporal boundaries on the GHG data and information that must be assessed. Criteria are the set of requirements against which the project is evaluated.

In determining the scope of the assessment, JCI takes into account the physical boundaries, sites or facilities of the project and the temporal boundaries (ie, the years when GHG emission reductions and removals are quantified). For verification, the temporal boundaries are determined by the length of the monitoring period.

The mandatory requirements of the VCS Program and ISO 14064-2 guide the criteria against which the verification is conducted. The methodology applied to the project also informs the criteria for verification; JCI thoroughly understands a methodology prior to undertaking an assessment. Where projects apply methodologies from CDM, JCI shall refer to any guidance provided by CDM with regard to the application of the methodology.

VVB is not expected to document every criterion that will apply to the verification engagement. Instead, it is sufficient to indicate the relevant documents containing the criteria such as the VCS Standard, ISO 14064-2 and the applied GHG methodology.

JCI’s verification is based on the requirements as defined under the VCS Program Guide /15/ and VCS Standard /16/, as well as the related VCS Guidance including the Validation and Verification Manual /18/.

1) JCI shall assess and verify that the implementation of the project activity and the steps taken to report emission reductions comply with the requirements as defined under the VCS Program Guide /15/ and VCS Standard /16/ and the relevant guidance provided by the VCS Association (VCSA).
2) This assessment shall involve a review of relevant documentation as well as a site visit in accordance with paragraph 3.3 of the Validation and Verification Manual /18/ of VCS Program. The information to be verified is described below.
3) The JCI’s verification of the project documentation provided by the Project proponent shall be based upon both quantitative and qualitative information on emission reductions.

Quantitative information comprises the reported numbers in the monitoring report submitted to JCI. Qualitative information comprises information on internal management controls, calculation procedures, procedure for transfer of data, frequency of emissions reports and review and internal audit of calculations.

4) In addition to the monitoring documentation provided by the Project proponents, JCI reviews :

(a) The registered VCS PD /1/ including the monitoring plan and the corresponding validation report /2/;
(b) The applied monitoring methodology;
(c) VCS Program Guide /15/, VCS Standard /16/ and Relevant VCS Guidance;
(d) Relevant decisions, clarifications and guidance of UNFCCC CDM;
(e) Any other information and references relevant to the project activity's resulting emission reductions (e.g., IPCC reports, data on electricity generation in the national grid or laboratory analysis and national regulations).

1.3 Level of assurance

The VCS Program requires a reasonable level of assurance in verification that GHG assertions are free of material errors, omissions and misrepresentations. This is non-negotiable between the project proponent and the VVB. In a reasonable level of assurance engagement, the JCI tests a sufficient amount of data to ensure with confidence that no material errors are present.

JCI conducts a verification of the project and summarises the verification results in the Verification Report which is based on the VCS Project Description (PD), the VCS Monitoring Report (MR), the supporting evidences which are available to the verifier, and the information collected through interview with the project proponent and site visit. JCI planed and organized to achieve a reasonable level of assurance of the verification conclusion.

1.4 Summary Description of the Project

48 MW Houqiao Hydropower Project, Yunnan, China (hereafter referred to as the project) is consists of three 16MW units, in total 48MW capacity of hydropower generation units developed at Binglang River in Tengchong County, Baoshan City, Yunnan Province, P. R. China. The objective of the project is to utilize water resources of the Binglang River for electricity generation, of which the PD was uploaded on the web site of VCD Program on 24th November, 2009. The project is a grid-connected renewable energy project, from which the electricity generated is sold to the Dehong Power Grid and then fed to Yunnan Power Grid, an integral part of the China Southern Power Grid (hereafter referred to as CSPG). The project activity contributes to the obvious GHG emission reductions by avoiding CO_2 emissions, as CSPG is dominated with fossil fuel fired power plants.

The monitoring period of this time is from 1st April, 2006 to 30th November, 2011. The total electricity replaced by the project during this monitoring period was 1,082,939,655MWh; therefore the emission reduction is 943,454 tCO2e throughout the monitoring period.

The PD has applied the following methodology and Tools of CDM to the project activity;

1) Methodology of ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources" (ver.10) /11/,
2) Tool for the demonstration and assessment of additionality (ver.05.2) /12/,
3) Tool to calculate project or leakage CO2 emissions from fossil fuel combustion (ver.02) /13/,

4) Tool to calculate the emission factor for an electricity system (ver.01.1) /14/

Table 1 Protocol (CL-1)

No. of request	CL-1
Clarification request	MR 1.8. Title and Reference of Methodology The version number of each Tool shall be shown in the Monitoring Report.
Reference	
Summary of project owner response	The version number of each Tool is added in the Monitoring Report.
Verification team conclusion	The Monitoring Report has revised appropriately. CL-1 was resolved and closed.

2 VALIDATION PROCESS, FINDINGS AND CONCLUSION

2.1 Validation Process

JCI shall apply standard auditing techniques to assess the quality of the information, including but not limited to:

(a) Desk review, involving:

(i) A review of the data and information presented to verify their completeness;
(ii) A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures;
(iii) An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

The documentations reviewed during the validation are same as the list in 3.2 Document Review.

(b) On-site assessment, involving:

(i) An assessment of the implementation and operation of the proposed VCS project activity as per the registered PD and the Monitoring Plan;
(ii) A review of information flows for generating, aggregating and reporting the monitoring parameters;
(iii) Interviews with relevant personnel to confirm that the operational and data collection procedures are implemented in accordance with the revised monitoring plan;
(iv) A cross-check between information provided in the monitoring report and data from other sources such as plant log books, inventories, purchase records or similar data sources;
(v) A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the PD and the selected methodology;
(vi) A review of calculations and assumptions made in determining the GHG data and emission reductions;
(vii) An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

The on-site visit and interview with stakeholders conducted for the validation are same as the list in 3.3 Interviews.

Table 2 Protocol (CL-2)

No. of request	CL-2
Clarification request	MR 2.1 Implementation Status of the Project Activity It is reported "Starting date of reducing GHG emissions". It is required to clarify it.
Reference	
Summary of project owner response	The statement "Starting date of reducing GHG emissions" is revised as "Starting date of the power generation".
Verification team conclusion	The Monitoring Report has revised appropriately. CL-2 was resolved and closed.

Table 3 Protocol (CL-3)

No. of request	CL-3
Clarification request	MR 3.2 Data and Parameters Monitored It is reported "Directly checked the plate by DOE during the onsite verification" in Cap_{PJ}. It should be revised since monitoring should be made by Project Proponent.
Reference	
Summary of project owner response	The statement "Directory checked the plate by DOE during the onsite verification" in the box of "Frequency of monitoring / recording" is revised as "Every on site verification".
Verification team conclusion	The Monitoring Report has revised appropriately. CL-3 was resolved and closed.

Table 4 Protocol (CL-4)

No. of request	CL-4
Clarification request	MR 3.3 Description of the Monitoring Plan The Monitoring Report shows that "Main meters M2, M3, M5 and M7 are ------; only meter M2 is ------ used as cross check meter". The description should be revised.
Reference	
Summary of project owner response	The statement "M2" in the relevant part is deleted and revised appropriately, i.e., "M3, M5 and M7 are ------".
Verification team conclusion	The Monitoring Report has revised appropriately. CL-4 was resolved and closed.

2.2 Validation Findings

2.2.1 Gap Validation

The project is only applied to and registered with the VSC Program and not to the CDM project and another approved GHG Program. JCI has confirmed it by desk review of the project documents and interview with the project proponent. So a gap validation is not required on the project.

2.2.2 Methodology Deviations

There is not any methodology deviation applied to the project. JCI has confirmed it by desk review of the project documents and information, interview with the project proponent and on site visit.

2.2.3 Project Description Deviations

There is one deviation in the monitoring plan from the registered PD dated 14th October, 2009 /1/. The deviation is that the applied national standard for electric meter calibration has changed from “Chinese Verification Regulation of Verification Equipment for AC Electrical Energy Meters, JJG597-2005” /38/ in the registered PD to “Chinese Verification Regulation of Electrical Energy Meters with Electronics, JJG596-1999” /39/. The PD author explained that it was a miss application of standard and the Yunnan Provincial Bureau of Quality and Technical Supervision, which made calibration works in the project, has advised the project proponent of its change. The detail of this change is reported in the section 2.2 Description from the Monitoring Plan.

VCS Validation and Verification Manual /18/ stipulates as follows regarding the project description deviation;

- Where the deviation impacts applicability of the methodology, additionality or appropriateness of the baseline scenario, the project proponent must describe and justify the deviation in a revised version of the project description. The requirement for a revised project description is in recognition of the deviation being a substantial change to the project.
- Where the deviation does not impact the applicability of the methodology, additionality or the appropriateness of the baseline scenario, and the project remains in compliance with the applied methodology, the project proponent must describe and justify the deviation in the monitoring report. The deviation is documented in the monitoring report in recognition of the deviation being a more limited change to the project.

In case of the deviation of this project, it is a miss application of standard in the monitoring plan. The deviation does not impact on accuracy of monitoring results.

The project proponent has described and justified the deviation in the monitoring report /4/ that it does not impact the applicability of the methodology, additionality or the appropriateness of the baseline scenario, and the project remains in compliance with the applied methodology.

JCI concluded that the deviation was acceptable in accordance with the VCS requirement since it was in compliance with the applied methodology. And also JCI confirmed that the project proponent described and justified the deviation in the monitoring report appropriately. JCI concluded that the deviation was documented in the monitoring report appropriately in recognition of the deviation being a more limited change to the project.

2.2.4 New Project Activity Instances

In the registered VSC PD /1/ the project is defined as a grouped project. During the monitoring period between 1st April 2006 and 30th November 2011 there was not any project inclusion to the project. So validation is not required regarding new project activity instance.

2.3 Validation Conclusion

JCI concluded that gap analysis was not required since there was not any methodology deviation and also was not the project description deviation to impact on applicability of methodology, and also there was not any project inclusion as new project activity instance.

3 VERIFICATION PROCESS

3.1 Method and Criteria

JCI shall apply standard auditing techniques to assess the quality of the information including but not limited to:

(a) Desk review, involving:

(i) A review of the data and information presented to verify their completeness;
(ii) A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures;
(iii) An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

(b) On-site assessment, involving:

(i) An assessment of the implementation and operation of the proposed VCS project activity as per the registered PD and the Monitoring Plan;
(ii) A review of information flows for generating, aggregating and reporting the monitoring parameters;
(iii) Interviews with relevant personnel to confirm that the operational and data collection procedures are implemented in accordance with the revised monitoring plan;
(iv) A cross-check between information provided in the monitoring report and data from other sources such as plant log books, inventories, purchase records or similar data sources;
(v) A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the PD and the selected methodology;
(vi) A review of calculations and assumptions made in determining the GHG data and emission reductions;

(vii) An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

3.2 Document Review

The following table outlines the documentation reviewed during the verification:

Table 5 Document list

No.	Title
PDD, Methodology, Tools, Guidance, Guidelines, and Manual	
/1/	Registered PD of “48 MW Houqiao Hydropower Project, Yunnan, China” (Version 1.3), dated 14/10/2009
/2/	Validation Report of “48 MW Houqiao Hydropower Project, Yunnan, China”, dated 15/10/2009, by JCI
/3/	Monitoring Report “48 MW Houqiao Hydropower Project, Yunnan, China” for 1st monitoring period (Version 01.1), dated 30/07/2012
/4/	Monitoring Report “48 MW Houqiao Hydropower Project, Yunnan, China” for 1st monitoring period (Version 03.0), dated 15/03/2013
/5/	Summary of On-site Assessment, dated 25/09/2012
/6/	Emission Reduction Calculation Sheet Houqiao, by Tepia Japan
Methodology, Tools, Guidance, Guidelines, and Manual	
/11/	ACM0002 “Consolidated baseline methodology for grid connected electricity generation from renewable sources” (Version 10)
/12/	Tool to demonstrating and assessment of additionality (Version 05.2)
/13/	Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion (Version 02)
/14/	Tool to calculate the emission factor for an electricity system (Version 01.1)
/15/	VCS Program Guide (Version 3.4), VCSA, 04/10/2012
/16/	VCS Standard (Version 3.3), VCSA, 04/10/2012
/17/	VCS Registration and Issuance process (Version 3.4), VCSA, 04/10/2012
/18/	VCS Validation and Verification manual (Version 3.0), VCSA, 04/10/2012
/19/	VCS Monitoring Report Template (Version 3.2), VCSA, 04/10/2012
/20/	VCS Verification Report Template (Version 3.2), VCSA, 04/10/2012
/21/	CDM Project Standard (version 02.1), CDM-EB

No.	Title
/22/	CDM Validation and Verification Standard (version 03.0), CDM-EB
/23/	VCS verification deed of representation by JCI, 13/03/2013
Monitoring data & CER calculation	
/31a/	Electricity invoice, 2011. 9
/31b/	Electricity invoice, 2011.10
/31c/	Electricity invoice, 2011.11
/31d/	Electricity invoice, 2011.12
/32/	Operating record, 2005-2011
/33/	Measurement result of the surface area of the reservoir, 2012.8, by Kunming Investigation, Design & Research Institute
/34a/	Calibration report – M 3, #121 (SN 0502A023-1), 11/08/2005
/34b/	Calibration report – M 3, #121 (SN 0502A023-1), 11/08/2010
/35a/	Calibration report – M 5, #321 (SN 0502A024-1), 11/08/2005
/35b/	Calibration report – M 5, #321 (SN 0502A024-1), 11/08/2010
/36a/	Calibration report – M 7, #322 (SN 0502A025-1), 11/08/2005
/36b/	Calibration report – M 7, #322 (SN 0502A025-1), 11/08/2010
/37/	Spread sheet for Monthly Invoice data (2006 – 2011)
/38/	Chinese Verification Regulation of Verification Equipment for AC Electrical Energy Meters, JJG597-2005,
/39/	Chinese Verification Regulation of Electrical Energy Meters with Electronics, JJG596-1999.
Others	
/41/	VCS Monitoring Management Manual, by Yunnan Baoshan Binglang River Hydropower Development Co., Ltd., 01/12/2009

No.	Title
/42a/	Operating Staff Training Record, 2009.12
/42b/	Operating Staff Training Record, 2010.12
/42c/	Operating Staff Training Record, 2011.12
/43/	Power Purchase Agreement, 25/07/2005, between Yunnan Baoshan Binglang River Hydropower Development Co., Ltd. and Yunnan Baoshan Grid Company
/44/	Power Purchase Agreement, 31/12/2010, between Yunnan Baoshan Binglang River Hydropower Development Co., Ltd. and Yunnan Baoshan Grid Company
/45/	Meter instruction manual
/46/	Photos of electrical meters
/47/	Technical administrative code of electric energy metering (DL/T 448-2000)

3.3 Interviews

The verification team visited the Project site on 24th to 25th September 2012 to verify whether the project has been implemented, monitored and reported as planned. During the visit, the team interviewed with relevant stakeholders listed in the below table with the topics discussed /5/.

Table 6 List of interviewees during on-site assessment

No.	Date	Name	Organization	Topics
/50/	24/09/2012	Mr. Ma Chuan Ms. Duan Penlian Mr. Yang Guoliang Ms. Wang Nan Ms. Keiko Tsuji Mr. Wang Shaohai	YBBRHDC Tepia Japan Sun Valley	**Site visit & Interview with Project Owner** • Confirmation of plant facilities and monitoring system • Project implementation status

No.	Date	Name	Organization	Topics
/51/	24-25 /09/2012	Mr. Ma Chuan Ms. Duan Penlian Mr. Yang Guoliang Ms. Wang Nan Ms. Keiko Tsuji Mr. Wang Shaohai	YBBRHDC Tepia Japan Sun Valley	**Interview with Project Owner** • Organization for monitoring, data handling, ER calculation and QA system • Calibration of monitoring devices • Evidence related to the monitoring **Review of 1st Periodic monitoring data** • Monitoring data • Calibration results of monitoring devices • ER calculation method **Interview with Tepia Japan** • Monitoring report • Calculation procedure of ER
/52/	25/09/2012	Mr. Ma Chuan Ms. Duan Penlian Mr. Yang Guoliang Ms. Wang Nan Ms. Keiko Tsuji Mr. Wang Shaohai	YBBRHDC Tepia Japan Sun Valley	**Interview with Project Owner, Tepia Japan** • Confirmation of evidence • Summary of on-site assessment

YBBRHDC; Yunnan Baoshan Binglang River Hydropower Development Co., Ltd.
Tepia Japan; Tepia Corporation Japan Co., Ltd.
Sun Valley; Yunnan Sun Valley Energy Conservation Industry Development Co., Ltd.

3.4 Site Inspections

The assessment team has conducted site inspection on 24/09/2012 and interviews with the project proponent to cross check the compliance of the project activity along with the project description document. The related documents of the project activity had been assessed to confirm the actual situation during the site visit.

The results of the site inspection are as follows;

1) It was confirmed by the name plates that the model of Turbine / Generator was consistent with the registered Project Description.
2) It was confirmed that the monitoring equipment have been maintained in consistence with the registered Monitoring Plan.

3) It was confirmed that the data of main meter was read by the operator twice a day at 08, 16 and 24 o'clock. Usually it was within +/- 5 minutes of the specified time.
4) The assessment team conducted the interview with the operator on training operation and VCS.

3.5 Resolution of Any Material Discrepancy

JCI, during its verification, shall identify issues related to the monitoring, implementation or operations of the proposed VCS project activity that could impair the capacity of the proposed VCS project activity to achieve emission reductions or influence the reporting of emission reductions. JCI shall identify, discuss and conclude these issues in the verification report.

JCI shall raise a CAR if one of the following occurs:

(a) Non-conformities with the monitoring plan or methodol
(b) ogy are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
(c) Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the Project proponents;
(d) Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
(e) Issues identified in a FAR during validation to be verified during verification have not been resolved by the Project proponents.

JCI shall raise a clarification request (CL) if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

JCI shall raise a FAR during verification for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

All CARs and CLs raised by the VVB during verification shall be resolved prior to submitting a request for issuance.

JCI shall report on all CARs, CLs and FARs in this verification report. This reporting shall be undertaken in a transparent manner that allows the reader to understand the nature of the issue raised the nature of the responses provided by the Project proponents, the means of verification of such responses and clear references to any resulting changes in the monitoring report or supporting annexes.

JCI shall resolve or "close out" CARs and CLs only if the Project proponents modify the monitoring report or provide adequate additional explanations or evidence that satisfies the JCI's concerns. If this is not done, JCI shall not recommend the project activity for VCUs issuance to the VCSA.

The verification protocol consists of the following description.

- ✧ Ref. to question in Protocol :
  Reference to the checklist question number in Protocol where the **CAR, CL** or **FAR** is explained.
- ✧ Summary of project owner response :
  The responses given by the project proponents during the communications with the verification team should be summarised in this section.
- ✧ Verification conclusion :
  This section should summarise the verification team's responses and final conclusions.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

The project was registered on 24th November, 2009 according to the information on the Project pipeline of the VCS website. The 1st crediting period of the project is from 01/04/2006 to 31/03/2016. The Validation Report of the project /2/ showed that all of CARs and CLs were closed in the Protocol.

The project is a 48 MW hydropower station located in the Binglang River, Tengchong County, Baoshan City, Yunnan Province, P. R. China. The installed capacity of the project is total 48MW (16MW x 3 units) with the expected annual grid supply electricity of 211,080 MWh (after 2009, and 149,514 MWh for 2006, 219,545 for 2007, and 187,980 for 2008, respectively).

The project started the construction on July 2003 and completed on December 2005. The commission operation started on September 2005. The monitoring of the VCS project started on 01/04/2006. The 1st monitoring period was from 01/04/2006 to 30/11/2011.

During on-site assessment for the Monitoring Report JCI confirmed that all physical features of the project described in the registered PD and the Monitoring Plan /1/ were in place, and the actual implementation of the project was in accordance with the registered PD including the Monitoring Plan /1/ and the methodology applied /11/.

The Monitoring Reports /3//4/ covers the monitoring period when it is from 01/04/2006 to 30/11/2011. During the monitoring period, the actual operation load factors of the plant (= Electricity generated in #1 to #3 units / Capacity / Days / 24hrs x100) are as follows;

* Electricity exported to the grid = EG_{ex} (M3 + M5 + M7) in the Monitoring Report

Table 7 Plant Load Factor

Period	Electricity generated (MWh)	Operation day (days)	Load Factor (%)
01/04/2006~31/12/2006	136,551,614	275	43.1
01/01/2007~31/12/2007	219,544,654	365	52.2
01/01/2008~31/12/2008	192,088,873	366	45.6
01/01/2009~31/12/2009	164,931,461	365	39.2
01/01/2010~31/12/2010	208,466,143	365	49.6
01/01/2011~30/11/2011	177,334,837	334	46.1
Total	1,098,897,580	2,070	46.1

In the monitoring period from 01/04/2006 to 30/11/2011, the average load factor is 46.1 % and the operation of the plant is relatively stable over the period. The average load factor is 55.8 % (=235,000MWh / 48MW / 8760hour) of expected in the Validation Report /2/. The actual load factor is somewhat lower than the expected figure but within assumption.

From above JCI concluded that such load factors reported in Table 7 are appropriate.

Table 8 Emission reductions

Item	Values applied in ex-ante calculation of the registered PD	Actual values reached during the monitoring period
Total Emission reductions *1 (tCO_2e)	1,137,915	943,454
Yearly Emission reductions (tCO_2e)	183,893	166,492 *2

*1; In the monitoring period from 01/04/2006 to 30/11/2011
*2; = 943,454/68 months*12 Months

JCI confirmed that all of information, which was provided in the monitoring report including monitored data and used variables, were reasonable and consistent with those in the registered PD.

4.2 Accuracy of GHG Emission Reduction or Removal Calculations

4.2.1 Monitoring equipment

As per the PD including the Monitoring Plan, the following monitoring equipments were installed in the proposed project plant.

Table 9 Monitoring Equipments

Item	Meter	Role	Place installed	Type	SN*2	Accuracy
EG_{ex} (M3)	M3	Electricity export from Project to Grid	Outlet of Project 110 kV line #121	ION8300*1	0502A023	0.2S
EG_{im} (M3)		Electricity import from Grid to Project				
EG_{ex} (M5)	M5	Electricity export from Project to Grid	Outlet of Project 35 kV line #321	ION8300*1	0502A024	0.2S
EG_{ex} (M7)	M7	Electricity export from Project to Grid	Outlet of Project 35 kV line #322	ION8300*1	0502A025	0.2S

*1; Bidirectional type meter *2; Serial Number

The project has two transmission lines connected with the Grid of which are 110 kV line and 35kV line.

There are two main maters which are shown in the Table 9. Those monitoring equipment are in consistence with the Monitoring Plan /1/ including accuracy level. There is not any back up meter for measurement of electricity export/import in the project site, but it can be backed up by the invoice issued by the Grid Company when the main meter is troubled or faulted.

JCI confirmed during on-site assessment that these meters were placed as specified in the revised PD including the Monitoring Plan /1/ and also in the Monitoring Report from 01/04/2006 to 30/11/2011 /4/.

The installed capacity (Cap_{PJ}) as monitoring item is confirmed as total 200 MW by on-site observation of the name plates of installed Turbine/Generator /4/.

The area of the reservoir (A_{pj}) as monitoring item is confirmed as 0.108 m^2 by the evidence measured by the design institute which is same as the value in the registered PD /4//33/.

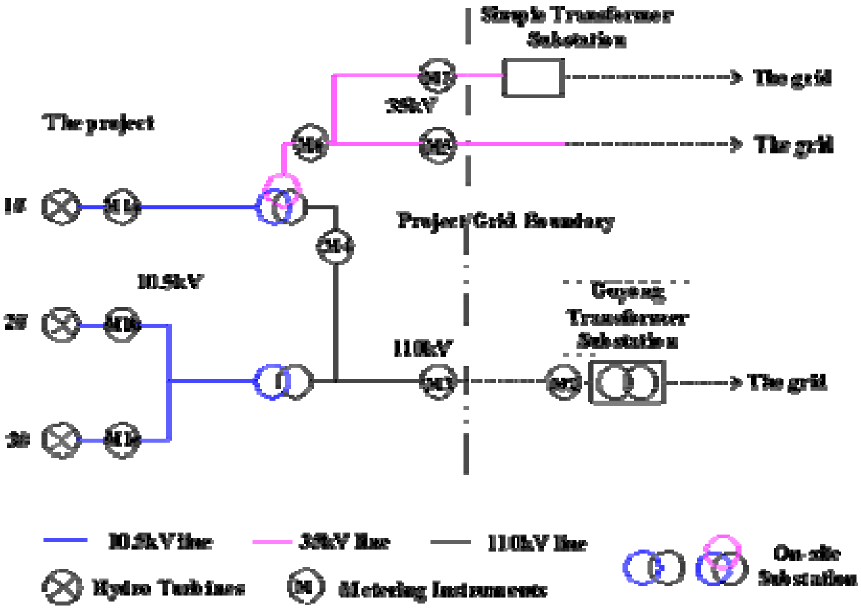

Figure 1 Electric line and measuring system

4.2.2 Calibration of measuring instruments

The PD including Monitoring Plan /1/ showed that measurement by electric meters was done according to the relevant national electric industry standard and regulations. It is interpreted that calibration of the project's electric meter shall be followed to those standard and regulations.

The project's meters, which should be calibrated, are M3, M5 and M7 measuring EG_{ex} and M3 measuring EG_{im} between the project and the Grid. The calibrations were based upon the codes of "Chinese Verification Regulation of Electrical Energy Meters with Electronics, JJG596-1999" in order to ensure the accuracy of the meters.

The Project proponent submitted the calibration records of all monitoring equipment /34a//34b/, /35a//35b/ and /36a//36b/ and those results were summarized as follows. All of meters used for monitoring were checked its accuracy by Yunnan Provincial Bureau of Quality and Technical Supervision. The calibrations were carried out before the start of commissioning and after 5 years in accordance with China code.

The calibration records showed that all of the main meters in the project have been calibrated as per requirements of the relevant national standards. The accuracy classes of meters for M3, M5 and M7 of EG_{ex} and M3 of EG_{im} are confirmed appropriately as 0.2S.

Table 10 Calibration results*1

No.	Itemde	SN	Accuracy Class	Date of Calibration	Result	On-line Period
M3	EG_{ex} (M3)	0502A023	0.2S	11/08/2005	Pass	01/04/2006 - 30/11/2011
	EG_{im} (M3)			11/08/2010	Pass	
M5	EG_{ex} (M5)	0502A024	0.2S	11/08/2005	Pass	01/04/2006 - 30/11/2011
				11/08/2010		
M7	EG_{ex} (M7)	0502A025	0.2S	11/08/2005	Pass	01/04/2006 - 30/11/2011
				11/08/2010		

*1; Calibrated by Yunnan Provincial Bureau of Quality and Technical Supervision

4.2.3 Baseline emission

The registered PD /1/ stated the baseline emission BE_y was calculated as follows in accordance with the applied methodology ACM0002 (Version 10) /11/ and "Tool to calculate the emission factor for an electricity system" (Version 01.1) /14/.

$$
\begin{aligned}
BE_y &= EG_y \times EF_{grid,CM,y} \\
&= (EG_{ex} - EG_{im}) \times EF_{grid,CM,y} \\
&= (EG_{ex}(M3 + M5 + M7) - EG_{im}(M3)) \times EF_{grid,CM,y}
\end{aligned}
$$

Where:

BE_y = Baseline emissions in year *y* (tCO_2/yr)

EG_y = Electricity supplied by the project activity to Grid (MWh).

$EF_{grid,CM,y}$ = Combined margin CO_2 emission factor for grid connected power generation in year y, which is calculated by the latest version of the "Tool to calculate the emission factor for an electricity system". The PD selected "Ex ante option" for emission factor determination and then the value of $EF_{grid,CM,y}$ was 0.8712 tCO_2/MWh from the PD.

EG_{ex} = The amount of electricity supplied by the proposed project to the grid (MWh).
EG_{ex} is calculated from the measurements of M3 (at 110 kV line), M5 and M7 (at 35 kV line).

EG_{im} = the amount of electricity get from the grid to the proposed project (MWh).
EG_{im} is the measurements of M3 (at 110kV line). There is no electricity import from the Grid through 35kV line.

Table 11-1 EG_{ex} Calculation

Year	EG_{ex} (Meter reading)				EG_{ex}	EG_y
	M3	M5	M7	M3+M5+M7	(Invoice)	For BE_y
	MWh	MWh	MWh	MWh	MWh	MWh
2006	132,835,592	2,764,510	931,510	136,531,612	136,553,204	136,531,612
2007	208,419,088	10,055,992	1,069,574	219,544,654	219,544,654	219,544,654
2008	174,807,370	15,960,242	1,321,261	192,088,873	190,949,632	190,949,632
2009	151,849,696	12,672,832	408,933	164,931,461	164,655,011	164,655,011
2010	199,555,938	8,564,384	345,821	208,466,143	208,466,143	208,466,143
2011	172,271,524	4,530,264	533,049	177,334,837	177,334,837	177,334,837
Total	1,039,739,208	54,548,224	4,610,148	1,098,897,580	1,097,503,482	1,097,481,889

Table 11-1 EG_{ex} Calculation

Year	EG_{ex}	EG_{im}		EG_y	$EF_{grid,CM,y}$	BEy
	From Table 11-2	M3 Reading	Invoice	$EG_{ex} – EG_{im}$		
	MWh	MWh	MWh	MWh	tCO_2/MWh	tCO_2/y
2006	136,531,612	1,178,202	1,178,202	135,353,410	0.8712	117,919
2007	219,544,654	1,929,716	1,929,716	217,614,938	0.8712	189,586
2008	190,949,632	4,109,358	4,109,358	186,840,274	0.8712	162,775
2009	164,655,011	3,859,676	3,859,676	160,795,335	0.8712	140,084
2010	208,466,143	2,570,388	2,570,388	205,895,755	0.8712	179,376
2011	177,334,837	894,895	894,895	176,439,942	0.8712	153,714
Total	1,097,481,889	14,542,235	14,542,235	1,082,939,655	0.8712	943,454

Table 11 shows the yearly data of EG_{ex}, EG_{im}, EG_y and BE_y. The EG_y total electricity supply from the project to Grid is 1,082,939,655 MWh for the monitoring period from 01/04/2006 to 30/11/2011.

JCI checked the monthly electricity exported to the Grid (EG_{ex}, $M3_{ex.}$ $M5_{ex,}$ $M7_{ex}$) and the monthly electricity imported from the Grid (EG_{im}, $M3_{im}$), reported respectively in Clause 5 of the Monitoring Report /4/. JCI concluded that the EG_y electricity supply from the project to Grid was selected lower value between the meter reading and the invoice and then it was conservative calculation of the baseline emission.

The baseline emission was calculated as follows;

$$BE_y = EG_y \times EF_{grid,CM,y}$$
$$= (EG_{ex} - EG_{im}) \times EF_{grid,CM,y}$$
$$= 943{,}454\ tCO_2e$$

JCI concluded that 943,454 tCO_2e in the Monitoring Report /4/ was calculated appropriately.

JCI verified the data and parameter used for the baseline emission calculation during the monitoring period from 01/04/2006 to 30/11/2011 and concluded that all of those data and parameters were correct.

Table 12 Protocol (CL-5)

No. of request	CL-5
Clarification request	MR 4.1 Baseline Emissions Table 1 shall be clarified whether its power supply data is conservative comparing with the invoice data. The table should show units of each item.
Reference	
Summary of project owner response	The conservative amount of power supply to the grid is reviewed and revised. For conservative calculation of BE_y, EG_{ex} and EG_{im} are selected lower value between meter reading and invoice.
	The Monitoring Report has revised Table 1 and added the detailed Table in Clause 5. It is clarified appropriately. CL-5 was resolved and closed.

4.2.4 Project emission

The reservoir area of the project is 0.17 m^2 and the power density is 282.4 W/m^2 in the registered PD /1/. The project proponent has submitted the evidence for the measured reservoir area at August 2012 which showed 0.17 m^2 as same as PD /33/. From the value the power density is calculated to be 282.4 W/m^2 which is more than 10 W/m^2 of the threshold of project emission in ACM0002 /11/. Then according to the applied methodology (ver10), the project emission should be neglected.

$$PE_y = 0\ tCO_2e.$$

Table 13 Protocol (CL-6)

No. of request	CL-6
Clarification request	MR 4.2 Project Emissions It is reported that "According to the validation report". It is an improper description. The project emission should be determined in accordance with Methodology or the registered PD.
Reference	
Summary of project owner response	The relevant description is revised as "According to the registered PD".
Verification team conclusion	The Monitoring Report has revised appropriately. CL-6 was resolved and closed.

4.2.5 Accuracy of Emission Reduction Calculation

The spread sheet of ER calculation /6/ was submitted and verified by JCI. The calculation results were shown in Table 1, 2 & 3 of the Monitoring Report.

Total baseline emissions (BE_y) : 943,454 tCO_2e

Total project emissions (PE_y) : 0 tCO_2e

Total leakage (L_y) : 0 tCO_2e

Total Emission Reductions (ER_y) : $ER_y = BE_y - PE_y - L_y$

$= 943{,}454$ tCO_2e

JCI checked the all calculation in the spread sheet of ER calculation /6/ and confirmed that there was no error in the spread sheet and also confirmed that there was no error, omission or misrepresentation in the related evidence and no discrepancy between evidence. So JCI concluded that the materiality of project was sufficiently lower than the five percent of the threshold for 'Project'. The project's Emission Reduction is 183,893 tCO_2e per year in the registered PD /1/ and then the project is categorized as 'Project' not 'Large project' in VCS Program.

Table 14 Protocol (CL-7)

No. of request	CL-7
Clarification request	MR 5. Additional Information This clause is not reported.
Reference	
Summary of project owner response	A table about the operating data from the monitoring period is added in the section 5, "Additional Information".
Verification team conclusion	The Monitoring Report has revised appropriately. CL-7 was resolved and closed.

4.3 Quality of Evidence to Determine GHG Emission Reductions or Removals

Through the verification process, JCI has used the relevant and reliable evidence listed in 3.2 Document Review of this report.

The evidences covering the monitoring period from 01/04/2006 to 30/11/2011 were provided by the project proponent and the VCS consultant, and JCI was able to validate the figures stated in the Monitoring Report /4/. All the measured monitoring data were obtained from the monitoring instruments appropriately calibrated and operated in accordance with the PD including Monitoring Plan and the National Regulations.

All of data and parameters used for the determination of the Emission Reduction ER_y are discussed in this Verification Report. Those data recorded and parameters are derived from the Monitoring Report /4/ and the registered PD including the Monitoring Plan /1/.

4.4 Management and Operational System

The project owner prepared the VCS Monitoring Management Manual /41/ to stipulate the organization and role of VCS management team, data collection process, monitoring device calibration, ER calculation, data and information preservation and so on.

JCI confirmed through the desk review and on-site assessment that the monitoring data and information management has been implemented consistent with the PD including the Monitoring Plan /1/ in accordance with the requirement of VCS Program.

Table 15 Protocol (CL-8)

No. of request	CL-8
Clarification request	MR 3.3 Description of the Monitoring Plan The VCS Monitoring Report Template requires reporting organizational structure, data handling procedure and so on. The Monitoring Report does not report such items.
Reference	
Summary of project owner response	The information about the reporting organizational structure and data handling procedure of the project owner is described.
Verification team conclusion	The Monitoring Report has revised appropriately to report the organization structure and so on. CL- 8 was resolved and closed.

5 VERIFICATION CONCLUSION

Japan Consulting Institute (JCI) has performed the verification of the project "48 MW Houqiao Hydropower Project, Yunnan, China", with regard to the relevant requirements of VCS Program. This verification covers the period from 01/04/2006 to 30/11/2011.

It is JCI's responsibility to express an independent verification statement on the reported GHG emissions and the GHG emission reductions both from the project activity for the period.

JCI's verification is based on the requirements as defined under the VCS Program Guide /15/ and VCS Standard /16/, as well as the related VCS Guidance including the Validation and Verification Manual /18/.

The verification was performed based on the following documentation:

1) Registered PD dated 14/10/2009 /1/.
2) Approved monitoring methodology ACM0002 (ver.10), "*Consolidated methodology for grid-connected electricity generation from renewable sources*" /11/.
3) Tool to calculate the emission factor for an electricity system, version 01.1 /14/
4) Monitoring Report version 01.1 dated 30/07/2012 /3/ and version 03.0 dated 15/03/2013 /4/.

The verification process covered the auditing techniques to assess the quality of the information including desk review and on-site assessment, and also the quality confirmation of evidence in accordance with VCS Validation and Verification manual (Version 3.0) /18/.

JCI has verified that the information provided by the Project proponents was actual and representative of current operation of the project activity, that it has been in accordance with the Monitoring Plan in the registered PD /1/ and the applied Methodology /11/, and that the emissions reduction achieved have been correctly determined.

In summary, it is JCI's opinion, as Validation/Verification body, that the information, described in the Monitoring Report version 03.0 /4/, meets all relevant VCS rules and requirements of the VCS Program. And also, based on the information confirmed and verified by JCI, it is JCI's opinion that 943,454 tonnes of CO_2 equivalents, indicated in the Monitoring Report version 03.0 /4/, is correctly determined as the Emission Reductions by the Project proponent during the period from 01 April 2006 to 30 November 2011. JCI thus requests the issuance of the VCUs as a VCS project activity.

Table 16 GHG Emission Reduction

GHG Emission Reductions or Removals	tCO_2e
Baseline Emissions	943,454
Project Emissions	0
Leakage	0
Net GHG emission reductions or removals	943,454